

Equilibrium refinements in extensive-form games

Trembling hand perfection

Normal-Form Perfect Equilibrium

Trembling hand in choosing a pure strategy. Fix a finite, normal-form game $\Gamma = \{S_i, u_i\}_{i=1}^I$.

Definition 1 A perturbation is a function $\epsilon : \bigcup_{i=1}^I S_i \rightarrow (0, 1)$ such that $\forall i \in I, \sum_{s_i \in S_i} \epsilon(s_i) < 1$.

Definition 2 Given Γ and a perturbation ϵ , define $\Gamma(\epsilon) = \{M_i(\epsilon), u_i\}_{i=1}^I$ where $M_i(\epsilon) = \{\mu \in M(S_i) : \forall s_i \in S_i, \mu(s_i) \geq \epsilon(s_i)\}$ and u_i is extended to $M(S_i)$.

$\Gamma(\epsilon)$ is the perturbed game with perturbation ϵ .

- $\Gamma(\epsilon)$ is well defined if $\forall i, M_i(\epsilon) \neq \emptyset$.

Example

- Consider a normal-form game with two players that we identify as Alice indexed by α , and Bob indexed by β : $\Gamma = [S_\alpha, S_\beta, u_\alpha, u_\beta]$,
 - a is a generic element of S_α , b is a generic element of S_β .
 - $u_\alpha : S_\alpha \times S_\beta \rightarrow \mathbb{R}$.
- Let S_α contain tree strategies, $S_\alpha = \{a_1, a_2, a_3\}$.
 - $M(S_\alpha) = \{p = (p_1, p_2, p_3) \in \mathbb{R}_+^3 : \sum_{i=1}^3 p_i = 1\}$ is the set of mixed strategies available to Alice.
 - $M(S_\beta) = \{q \in \mathbb{R}_+^N : \sum_{i=1}^N q_i = 1\}$ is the set of mixed strategies available to Bob.

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- Extend the payoff function to mixed strategies,
 $(\forall b \in S_\beta) (\forall p \in M(S_\alpha)),$

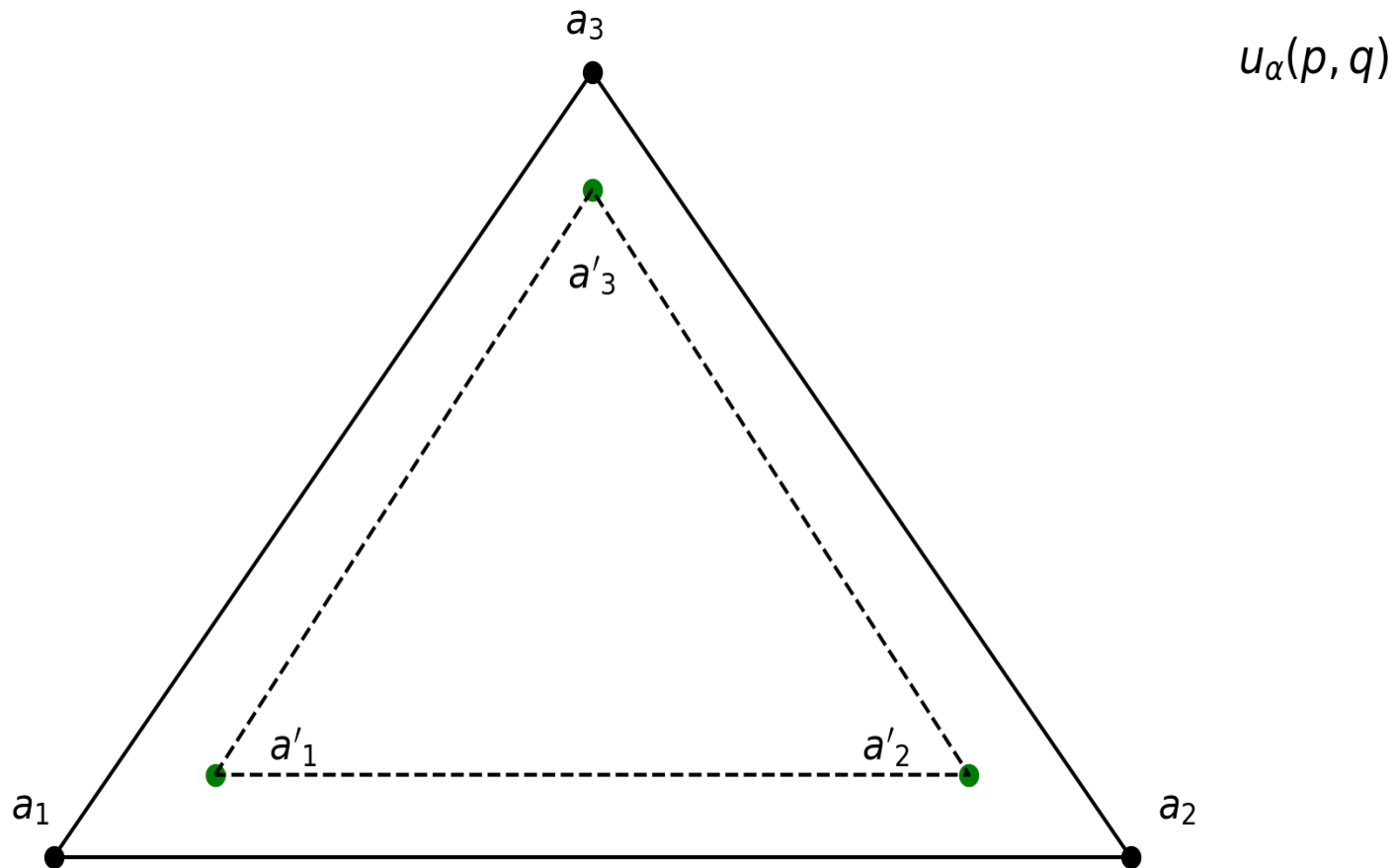
$$u_\alpha(p, b) = u_\alpha(a_1, b)p_1 + u_\alpha(a_2, b)p_2 + u_\alpha(a_3, b)p_3$$

$$u_\alpha(p, b) = \sum_{i=1}^3 u_\alpha(a_i, b)p_i$$

- Similar arguments applied to Bob yield

$$u_\alpha(p, q) = \sum_{a \in S_\alpha} \sum_{b \in S_\beta} u_\alpha(a, b)p_a q_b$$

Example



- $u_\alpha(p, b)$ is restricted to p in the smaller triangle; for $i = 1, 2, 3$, $p_i \geq \epsilon(a_i) > 0$.

$$\begin{aligned}
a_1 &\sim \overbrace{(1,0,0)}^p \rightarrow a'_1 \sim (1 - \epsilon(a_1) - \epsilon(a_2), \epsilon(a_2), \epsilon(a_3)) \\
a_2 &\sim (0, 1, 0) \rightarrow a'_2 \sim (\epsilon(a_1), 1 - \epsilon(a_1) - \epsilon(a_3), \epsilon(3)) \\
a_3 &\sim (0, 0, 1) \rightarrow a'_3 \sim (\epsilon(a_1), \epsilon(a_2), 1 - \epsilon(a_1) - \epsilon(a_2))
\end{aligned}$$

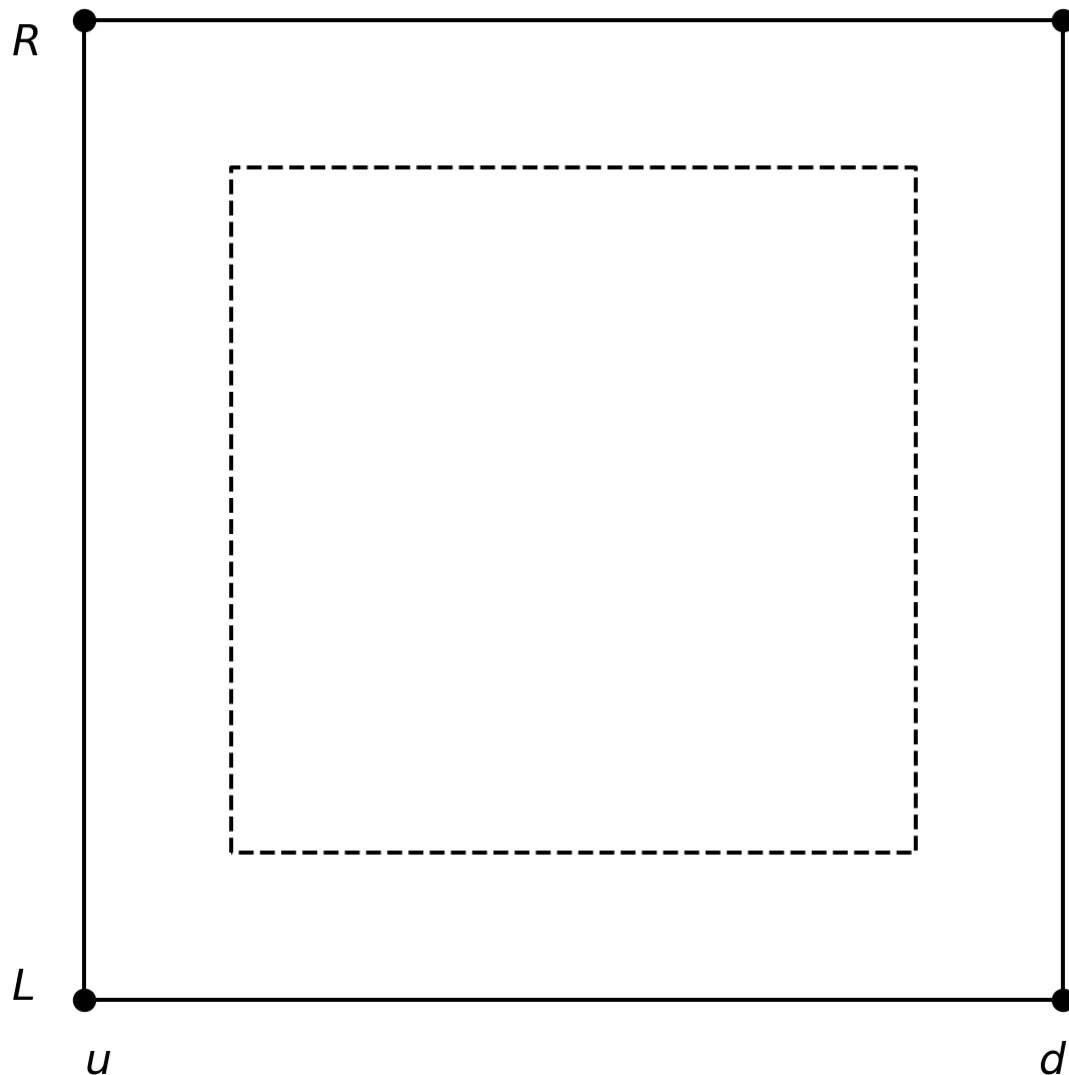
Example. Two-player, two-strategy game

	L	R	
u	1, 1	0, 3	$(1 - p_1)$
d	-3, 0	0, 0	(p_1)
	$(1 - q_1)$	q_1	

- Two NE, (u, L) and (d, R) . Let the perturbation be

Strategy	$\epsilon(\cdot)$
u	1/8
d	1/4
L	1/16
R	1/32

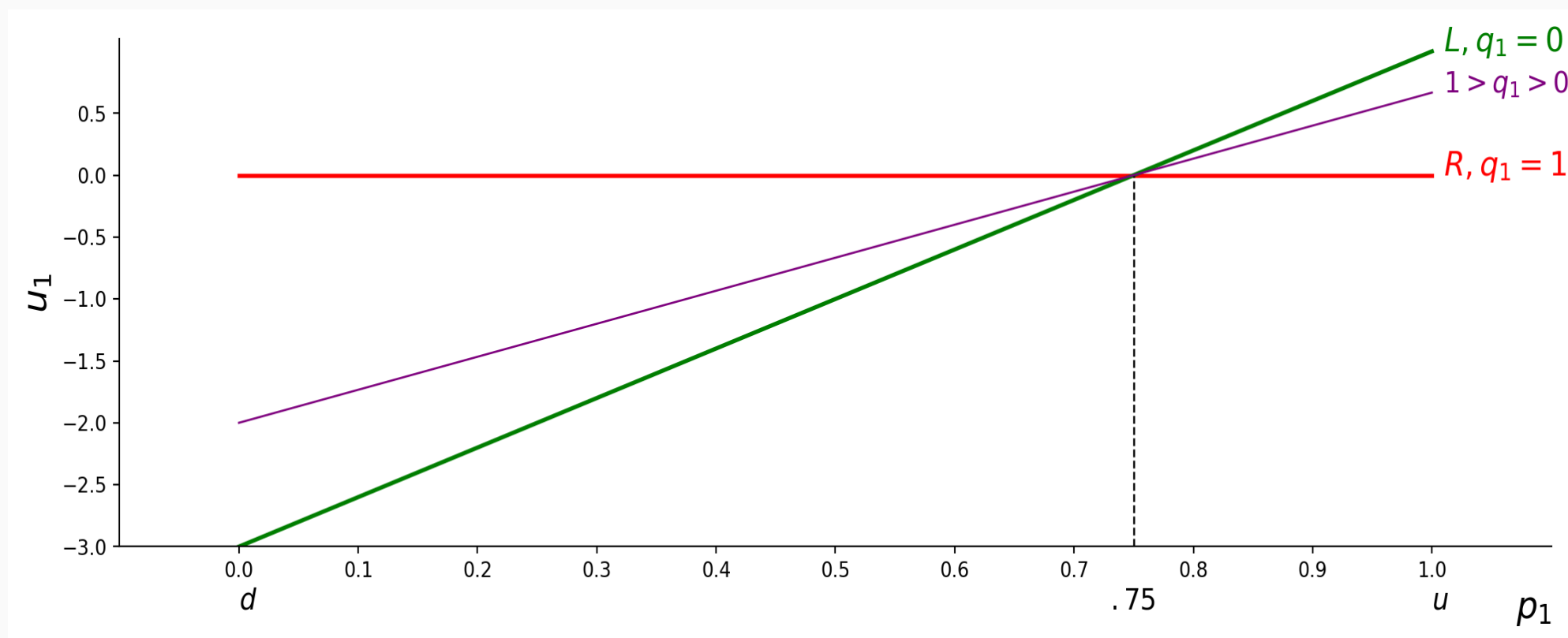
Relevant mixed strategies for $\Gamma(\epsilon)$



Example: Player 1's best response

$$u_1(p, R) = (1 - p_1) \times 0 + p_1 \times 0$$

$$u_1(p, L) = (1 - p_1) \times -3 + p_1 \times 1$$



Normal-form perfect equilibrium

Definition 3 (NFPE) Let $\Gamma = \{S_i, u_i\}_{i=1}^I$ be a finite normal-form game. A strategy profile μ is a Normal-Form Perfect Equilibrium (NFPE) if there is a sequence of perturbations $\epsilon^n : \|\epsilon^n\|_\infty \rightarrow 0$ and a NE μ^n of $\Gamma(\epsilon^n)$ such that $\lim_{n \rightarrow \infty} \mu^n = \mu$.

Example

	L	R	
u	1, 1	0, 3	$(1 - p_1)$
d	-3, 0	0, 0	(p_1)
	$(1 - q_1)$	q_1	

(d, R) is NE but is not NFPE.

- (d, R) in Γ is

$$\bar{\mu}_1(u) = 0 = (1 - p_1)$$

$$\bar{\mu}_2(L) = 0 = (1 - q_1)$$

Denote $\bar{\mu}_1$ as $(0, 1)$, $\bar{\mu}_2$ as $(0, 1)$.

- Let ϵ^n be any sequence of perturbations, $\|\epsilon^n\|_\infty \rightarrow 0$.

- Let $\mu^n = (\mu_1^n, \mu_2^n)$ be strategies in $\Gamma(\epsilon^n)$, $\mu_i^n(s_i) > \epsilon^n(s_i) > 0$.
- Let $\mu^n \rightarrow \bar{\mu} = [(0, 1), (0, 1)]$.

$$E[U_1|u, \mu_2^n] = \mu_2^n(L) \times 1 + \mu_2^n(R) \times 0$$

$$E[U_1|d, \mu_2^n] = \mu_2^n(L) \times (-3) + \mu_2^n(R) \times 0$$

- $\forall \mu_2^n(L) > 0, E[U_1|u, \mu_2^n] > E[U_1|d, \mu_2^n]$
- Then if $\mu_1^n \in BR(\mu_2^n)$, $\mu_1^n = (1, 0) \nrightarrow (0, 1)$

Some properties of normal-form PE

NFPE is commonly named Perfect Equilibrium.

Every NFPE is a NE.

Theorem 1 Every finite normal-form game has a NFPE.

Corollary 1 Every finite extensive-form game has a NFPE of its normal-form representation.

Theorem 2 Every NE in completely mixed strategies is a NFPE.

Lemma 1 A NFPE does not use weakly dominated strategies.

Lemma 2 A strategy profile μ in a two person finite normal-form game is a NFPE if it is a NE and for $i = 1, 2$, μ_i puts no mass on any weakly dominated strategy.

Example 3-player game

- Player 1's action space is $\{T, B\}$; Player 2's is $\{L, R\}$.
- Player 3 chooses "the game," her action space is $\{\ell, r\}$.

		L	R			L	R
3.ℓ	T	1,1,1	1,0,1	3.r	T	1,1,0	0,0,0
	B	1,1,1	0,0,1		B	0,1,0	1,0,0

- (B, L, ℓ) is a NE in undominated strategies; it is not NFPE.
- $2.L, 3.\ell, 1.B$ are undominated trivially. The profile is a NE; deviating will decrease the respective players' payoffs.
- *Exercise:* Is (T, L, ℓ) a NE? Is it a NFPE?

(B, L, ℓ) is not a NFPE

- Let μ_2^n, μ_3^n be any pair of completely mixed strategies such that

$$(\mu_2^n(L), \mu_2^n(R)) \rightarrow (1, 0) \quad L \text{ in the limit}$$

$$(\mu_3^n(\ell), \mu_3^n(r)) \rightarrow (1, 0) \quad \ell \text{ in the limit}$$

$$\begin{aligned}
& E[U_1|T, \mu_2^n, \mu_3^n] - E[U_1|B, \mu_2^n, \mu_3^n] \\
&= \sum_{s_2, s_3} [U_1(T, s_2, s_3) - U_1(B, s_2, s_3)] \mu_2^n(s_2) \mu_3^n(s_3) \\
&= (1 - 1) \mu_2^n(L) \mu_3^n(\ell) + (1 - 0) \mu_2^n(L) \mu_3^n(r) \\
&\quad + (1 - 0) \mu_2^n(R) \mu_3^n(\ell) + (0 - 1) \mu_2^n(R) \mu_3^n(r) \\
&= \mu_2^n(R) \mu_3^n(\ell) - \mu_2^n(R) \mu_3^n(r) + \mu_2^n(L) \mu_3^n(r) \\
&= \mu_2^n(R) [\mu_3^n(\ell) - \mu_3^n(r)] + \mu_2^n(L) \mu_3^n(r) > 0
\end{aligned}$$

because $\forall \{(\mu_2^n, \mu_3^n)\}_{n \geq 1}, \exists N : n > N \implies \mu_3^n(\ell) - \mu_3^n(r) > 0$.

		L	R			L	R
3.ℓ	T	1,1,1	1,0,1	3.r	T	1,1,0	0,0,0
	B	1,1,1	0,0,1		B	0,1,0	1,0,0

Extensive-form Perfect Equilibrium

- Trembles (mistakes) in different information sets are independent. Fix an extensive form game

$$\Gamma = [\mathcal{I}, (X, <), \{\mathcal{P}_i\}_{i=0}^I, \{\mathcal{U}_i\}_{i=1}^I, \{\mathcal{C}_h\}_{h \in \mathcal{U}_i, i \geq 1}, \{P_x\}_{x \in \mathcal{P}_0}, V]$$

Definition 4 A function $\epsilon : \bigcup_{h \in \bigcup \mathcal{U}_i} \mathcal{C}_h \rightarrow (0, 1)$ is perturbation of Γ if

$$(\forall i \in I), (\forall h \in \mathcal{U}_i), \sum_{a \in \mathcal{C}_h} \epsilon(a) < 1.$$

Definition 5 Given an extensive-form game Γ and a perturbation ϵ , $\Gamma(\epsilon)$ is the extensive form game Γ modified so that behavior strategies are limited to $\Sigma_i(\epsilon) = \{\sigma_i \in \Sigma_i : \forall h \in \mathcal{U}_i, \forall a \in \mathcal{C}_h, \sigma_i(a|h) \geq \epsilon(a)\}$.

$\Gamma(\epsilon)$ is called the perturbed game with perturbation ϵ .

- $\Gamma(\epsilon)$ is well defined if $\forall i, \Sigma_i(\epsilon) \neq \emptyset$.

Extensive-form Perfect Equilibrium

Definition 6 (EFPE) Let Γ be a finite extensive-form game. A behavior-strategy profile σ is an Extensive-Form Perfect Equilibrium (EFPE) if there is a sequence of perturbations ϵ^n such that $\|\epsilon^n\|_\infty \rightarrow 0$ and a NE σ^n of $\Gamma(\epsilon^n)$ such that $\lim_{n \rightarrow \infty} \sigma^n = \sigma$.

Theorem 3 Every EFPE of Γ is a SPE of Γ .

Theorem 4 Every finite extensive-form game with perfect recall has an extensive-form PE.

Comments

NFPE and EFPE differ from each other.

- An NFPE is a profile of distributions over pure strategies, an EFPE is a profile of functions mapping information sets to distributions over choices.
- An EFPE may fail to be NFPE (Exercise 7.36 in MSZ). An NFPE may fail to be EFPE (Exercise 7.37 MSZ).
- In EF games where each player has a single information set both NFPE and EFPE coincide.

References